

Soyeon Hong

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Profile

Machine learning researcher specializing in multimodal learning and LLM evaluation. I take a data-driven approach to modeling across diverse domains, including multimodal, vision, and materials science data. My research focuses on building rigorous, reproducible pipelines that measure real model capability.

Education

Ph.D. in Artificial Intelligence

2022 – Present

Ajou University, Suwon, Republic of Korea

Advisor: Prof. Hyunsouk Cho

Research focus: multimodal representation learning, vision-language models, and LLM evaluation.

B.S. in Psychology and Software & Computer Engineering

2017 – 2022

Ajou University, Suwon, Republic of Korea

Double major in Psychology and Software & Computer Engineering.

Experience

Research Assistant

Sep 2023 – Dec 2023

Kakao Mobility

- Built a LangChain-based RAG pipeline to support LLM-assisted writing for the company's annual report.
- Developed an evaluation system to assess the quality and stylistic consistency of LLM-generated writing.
- Implemented a web interface with Flask & Python to integrate the generation and evaluation workflow.

Publications

* denotes equal contribution.

- [1] **Soyeon Hong**, Jinchan Kim, Jaegook You, Seungtaek Choi, Suha Kwak, Hyunsouk Cho. "TextME: Bridging Unseen Modalities Through Text Descriptions." *ICML*, 2026.
- [2] Ji Hoon Hong, Chi-Hoon Lee, Hyeon Woo Kim, Dong Won Jeon, Jun Hyuk Kang, Yoon-Seo Kim, Ji-Su Bae, Seonggil Heo, Sangyeon Lee, Jooyeong Na, Jaegook You, **Soyeon Hong**, Hyunsouk Cho, Jin-Seong Park, Sung Beom Cho. "AI-driven Quantitative Review of Mobility-Stability Trade-off in Oxide Semiconductors." *Nano Convergence*, 2026.
- [3] Wansik Jo, Jooyeong Na, **Soyeon Hong**, Seungtaek Choi, Hyunsouk Cho. "Rethinking the Training Paradigm of Discrete Token-Based Multimodal LLMs: An Analysis of Text-Centric Bias." *CIKM*, 2025.
- [4] Jin-Hyuk Jeong*, **Soyeon Hong***, Ji-Hyeon Kwon, Soyoun Yang, Dong-Wook Kim, Nan Song, Hyunsouk Cho, Chun-Woong Park. "Machine Learning-Based Prediction of Aerodynamic Performance in Arformoterol-Lactose Dry Powder Inhaler Formulations Using Surface Roughness Features." *International Journal of Pharmaceutics*, 2025.
- [5] Sooho Kim*, **Soyeon Hong***, Kyungsoo Park, Hyunsouk Cho, Kyung-Ah Sohn. "Omnistitch: Depth-Aware Stitching Framework for Omnidirectional Vision with Multiple Cameras." *ACM MM*, 2024.
- [6] **Soyeon Hong**, Hyeoungguk Kang, Hyunsouk Cho. "Cross-Modal Dynamic Transfer Learning for Multimodal Emotion Recognition." *IEEE Access*, 2024.
- [7] **Soyeon Hong**, Jeonghoon Kim, Donghoon Lee, Hyunsouk Cho. "DeMuSA: Demo for Multimodal Sentiment Analysis." *ICME Workshop*, 2022.

AWARDS & HONORS

First Prize, 2026 AI Co-Scientist Challenge Korea — Ministry of Science and ICT (MSIT), Republic of Korea	Apr 2026
Outstanding Paper Award — Ajou University	Jan 2025
3rd Place, 2023 Kakao Mobility AI Hackathon — Kakao Mobility	Nov 2023
Industry-Academia Scholarship — POSCO DX	Jul 2023 – Dec 2023
First Prize, PaaS-TA Open Cloud Platform Idea Competition — Ministry of Science and ICT (MSIT)	Dec 2020
Encouragement Award, Commercial District Big Data Analysis Contest — Seoul Credit Guarantee Foundation	Mar 2020

RESEARCH PROJECTS

Ferroelectric Thin-Film Materials for High-Performance Semiconductor Devices via Data-Driven AI Design Jul 2024 – Present

National Strategic Materials R&D Project, IITP • Research Participant

- Contributed to AI-driven design of ferroelectric thin-film materials using a data-feedback framework.

Large Language Intelligence for Oxide Semiconductor Mobility-Stability Trade-off Apr 2024 – Jan 2026

Materials Global Young Connect Project, IITP • Research Participant

- Contributed to developing large language model techniques for materials science applications.

NCSOFT Industrial Project May 2023 – Sep 2023

iKnowLab, Ajou University • with NCSOFT • Research Participant

- Developed AI trading models for a constrained optimization problem using generative models.
- Ran model simulations and coordinated the project team.

Omni-Data-Driven Abductive Reasoning Framework for Complex Causal Understanding Apr 2022 – Dec 2026

National R&D Project (Human-Centered AI Core Technology), IITP • Research Participant

- Contributed to an abductive reasoning framework for understanding complex causal relationships from omni-modal data.

SKILLS

Python, C/C++, R, PyTorch, Hugging Face, DeepSpeed, vLLM, Docker

Languages: Korean (Native), English (Conversational)

TEACHING

Teaching Assistant • Ajou University

- Introduction to Open Source Software 2023, 2024–2025
- Computer Networks 2024
- Database 2022
- Natural Language Processing: Deep Learning Theory and Practice, KIBME Aug 2022

Tutor • Ajou University

- Korean Academic Writing Tutor for International Students, Global College 2026